

HTML Basics

GXEST203

Foundations of Computing: From Hardware Essentials to Web Design

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HTML Basics

What is HTML?

- HTML stands for **HyperText Markup Language**.
- It is the standard markup language for creating **web pages**.
- *HTML is not a programming language. It's a markup language used to create web pages.*
- It describes the **structure** of a web page.
- HTML elements are represented by **tags**.
- Tags can have **attributes** that provide additional information.
- Web browsers read HTML documents and render them as web pages.

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Basic Structure of an HTML Document

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Page Title</title>
5 </head>
6 <body>
7   <h1>This is a Heading</h1>
8   <p>This is a paragraph.</p>
9 </body>
10</html>
```

- `<!DOCTYPE html>`: Declares the document type.
- `<html>`: The root element of the page.
- `<head>`: Contains meta information about the document.
- `<body>`: Contains the visible content.

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HTML Basics

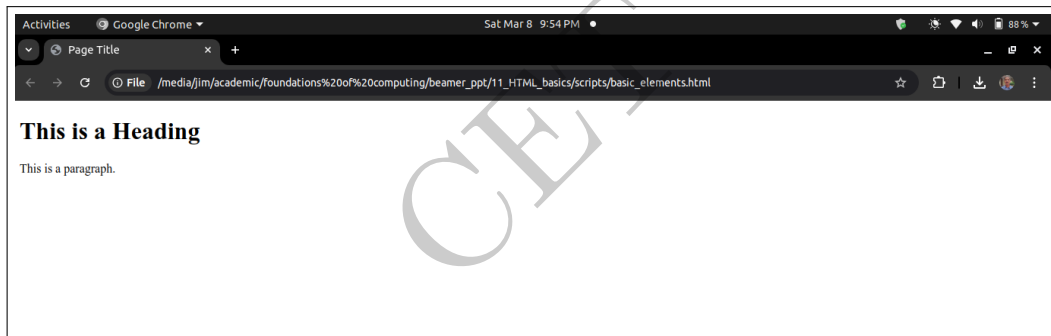


Figure: The page rendered by the above html code

HTML Elements

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <!-- Page metadata -->
5   <meta charset="UTF-8">
6   <meta name="viewport" content="width=device-width,initial-scale=1.0">
7   <meta name="description" content="Simple_HTML5_website.">
8   <title>My Website</title>
9 </head>
10 <body>
11   <!-- Page header -->
12   <header><h1>Welcome to My Website</h1></header>
13   <!-- Navigation bar -->
14   <nav>
15     <ul>
16       <li><a href="#home">Home</a></li>
17     </ul>
18   </nav>
19   <!-- Main content -->
20   <main>
21     <h2>About HTML</h2>
22     <p>HTML stands for <strong>HyperText Markup Language</strong>.</p>
23   </main>
24   <!-- Page footer -->
25   <footer><p>&copy; 2025 My Website.</p></footer>
26 </body>
27 </html>
```



Figure: The page rendered by the above html code

Page Meta data

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <!-- Page metadata: Provides information about the web page -->
5   <meta charset="UTF-8"> <!-- Character encoding for universal text support -->
6   <meta name="viewport" content="width=device-width,initial-scale=1.0"> <!--
7     Responsive design for mobile devices -->
8   <meta name="description" content="Simple HTML5 website."> <!-- Brief
9     description for search engines -->
10  <title>My Website</title> <!-- Title displayed on the browser tab -->
11 </head>
12 </html>
```

HTML Elements

- An **element** in HTML usually consists of a **start tag**, **content**, and an **end tag**. `< start tag > content < /end tag >`
- Some elements are **self-closing** (do not require an end tag)

Common HTML Tags

- `<h1>` to `<h6>`: Headings
- `<p>`: Paragraph
- `<a>`: Anchor (Hyperlink)
- `<img>`: Image
- `<ul>`, `<ol>`, `<li>`: Lists
- `<div>`: Division or section
- `<span>`: Inline container

Self-closing HTML Tags

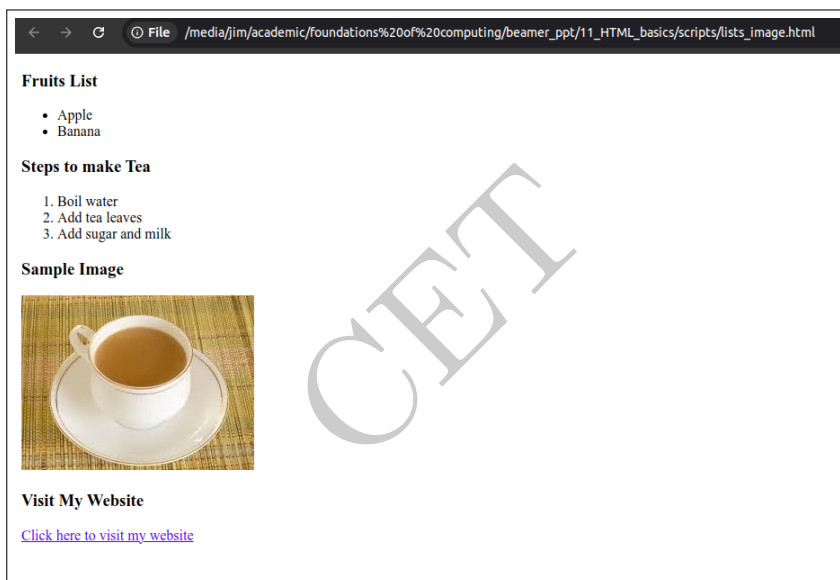
- `<br>`: Inserts a line break.
- `<hr>`: Creates a horizontal rule.
- `<img>`: Embeds an image.
- `<input>`: Creates an input field.
- `<link>`: Links to an external stylesheet.
- `<meta>`: Provides metadata about the page.
- `<param>`: Defines parameters for an object.

HTML Elements

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <title>HTML5 Example</title>
5 </head>
6 <body>
7   <!-- Unordered List Example -->
8   <h3>Fruits List</h3>
9   <ul>
10    <li>Apple</li>
11    <li>Banana</li>
12  </ul>
13   <!-- Ordered List Example -->
14   <h3>Steps to make Tea</h3>
15   <ol>
16    <li>Boil water</li>
17    <li>Add tea leaves</li>
18    <li>Add sugar and milk</li>
19  </ol>
20   <!-- Image Example -->
21   <h3>Sample Image</h3>
22   
23   <!-- Hyperlink Example -->
24   <h3>Visit My Website</h3>
25   <a href="https://www.example.com" target="_blank">Click here to visit my website</a>
26 </body>
27 </html>
```

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HTML Tables

```
1 <!DOCTYPE html>
2 <html lang="en">
3
4 <body>
5   <table border="1">
6     <!-- Table Header -->
7     <tr>
8       <th>Name</th>
9       <th>Age</th>
10    </tr>
11
12    <!-- First Row -->
13    <tr>
14      <td>Alice</td>
15      <td>25</td>
16    </tr>
17
18    <!-- Second Row -->
19    <tr>
20      <td>Bob</td>
21      <td>30</td>
22    </tr>
23  </table>
24 </body>
25 </html>
```

4

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HTML Basics

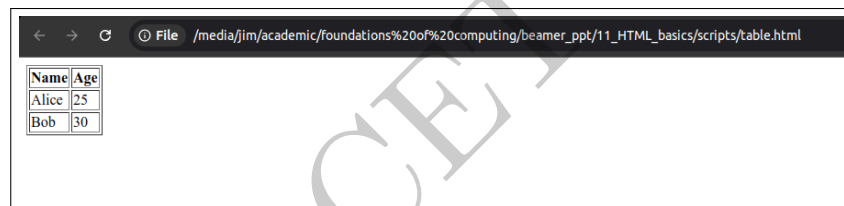


Figure: The page rendered by the above html code

Further Study Resources on HTML

- **MDN Web Docs by Mozilla**
<https://developer.mozilla.org/en-US/docs/Web>
- **w3schools.com**
<https://www.w3schools.com/html/default.asp>